

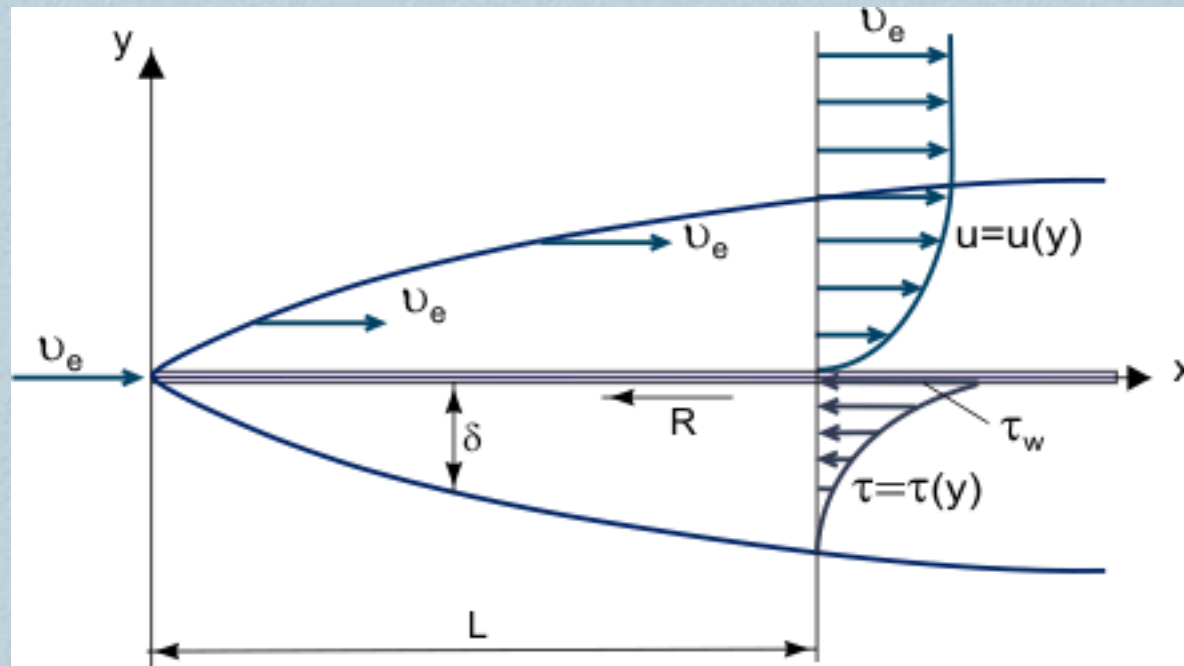
CREATE Module: Perturbation Theory Lecture 7: Boundary Layers (formal)

Marek Stastna

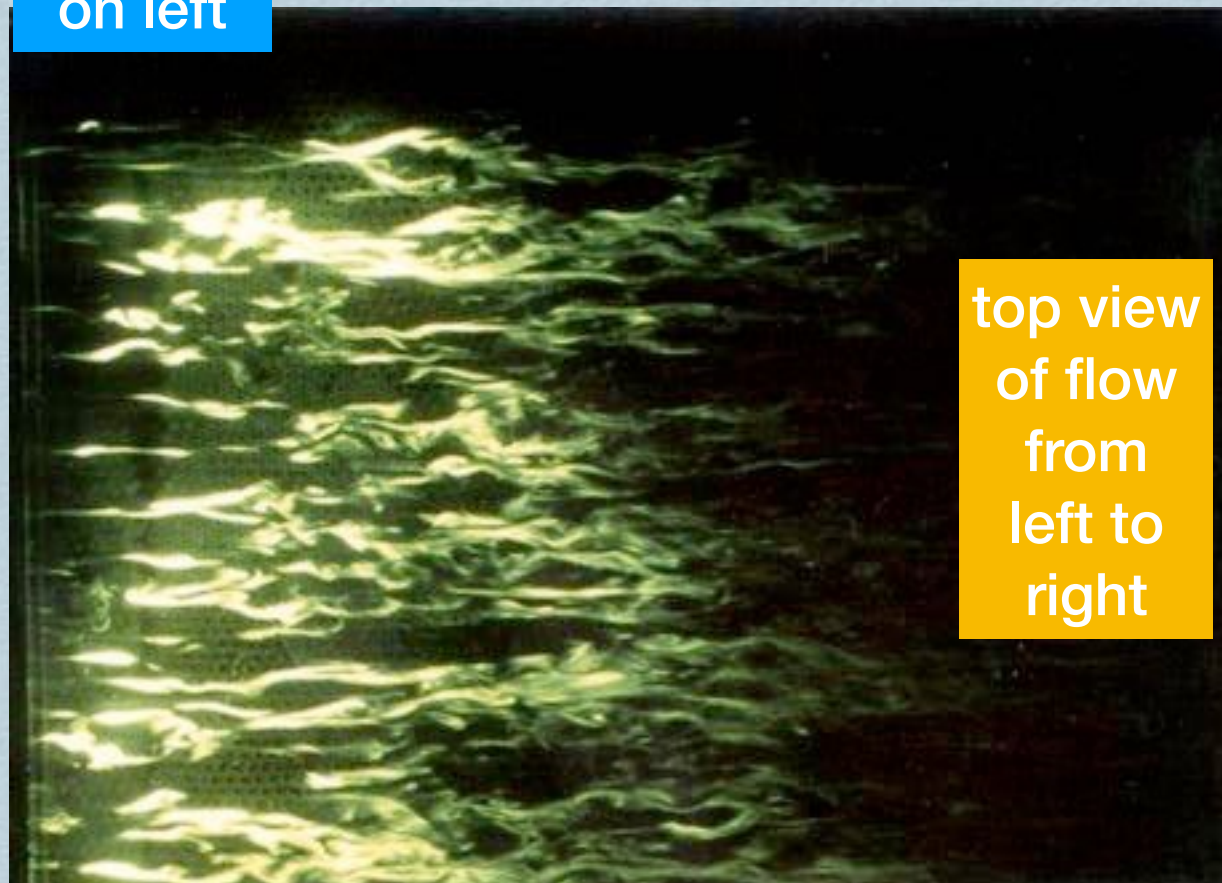
I can state flatly that heavier than air flying machines are impossible.

— *Lord Kelvin, 1895*

<https://www.youtube.com/watch?v=e1TbkLIDWys>

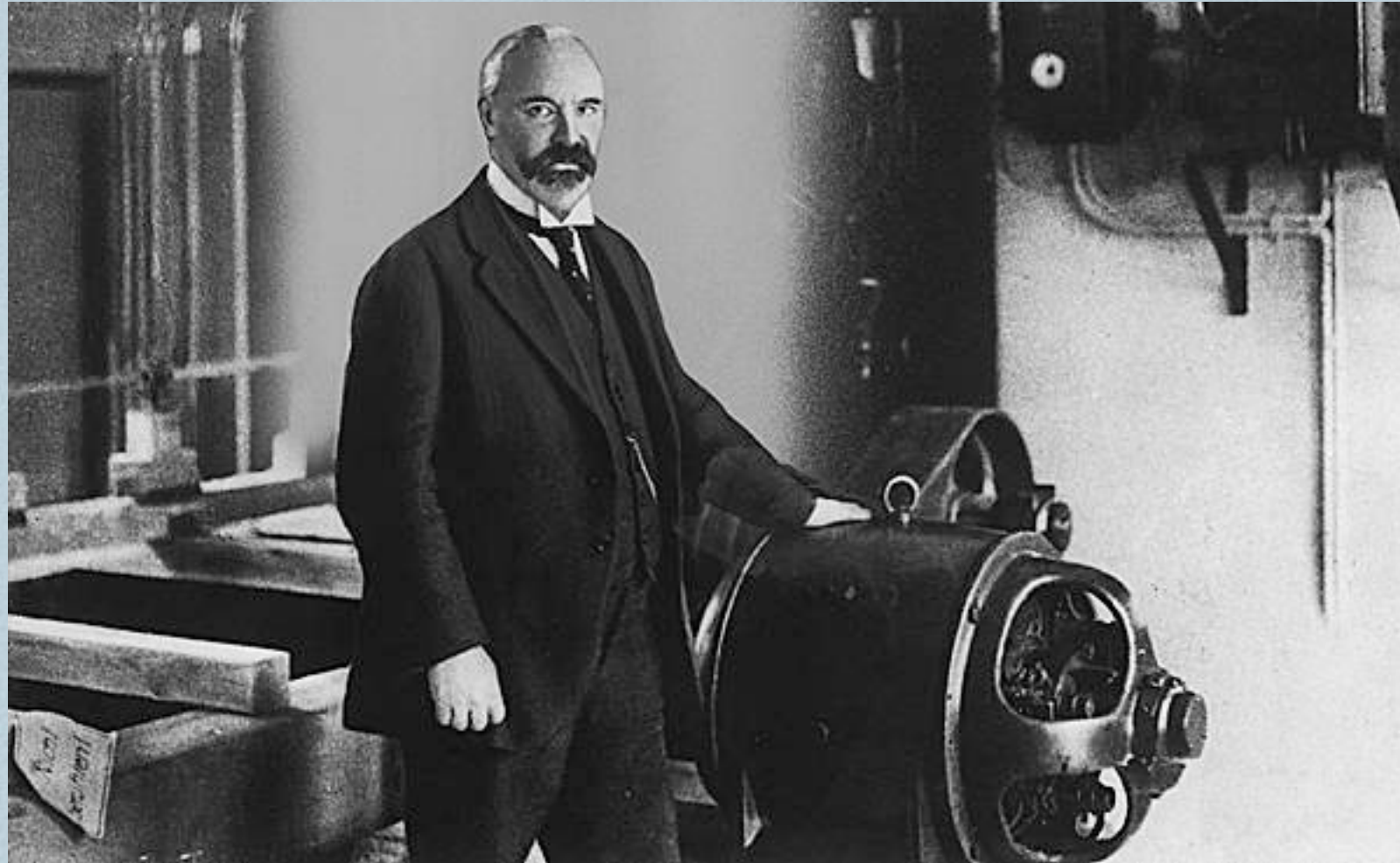


wire to
create
bubbles
on left



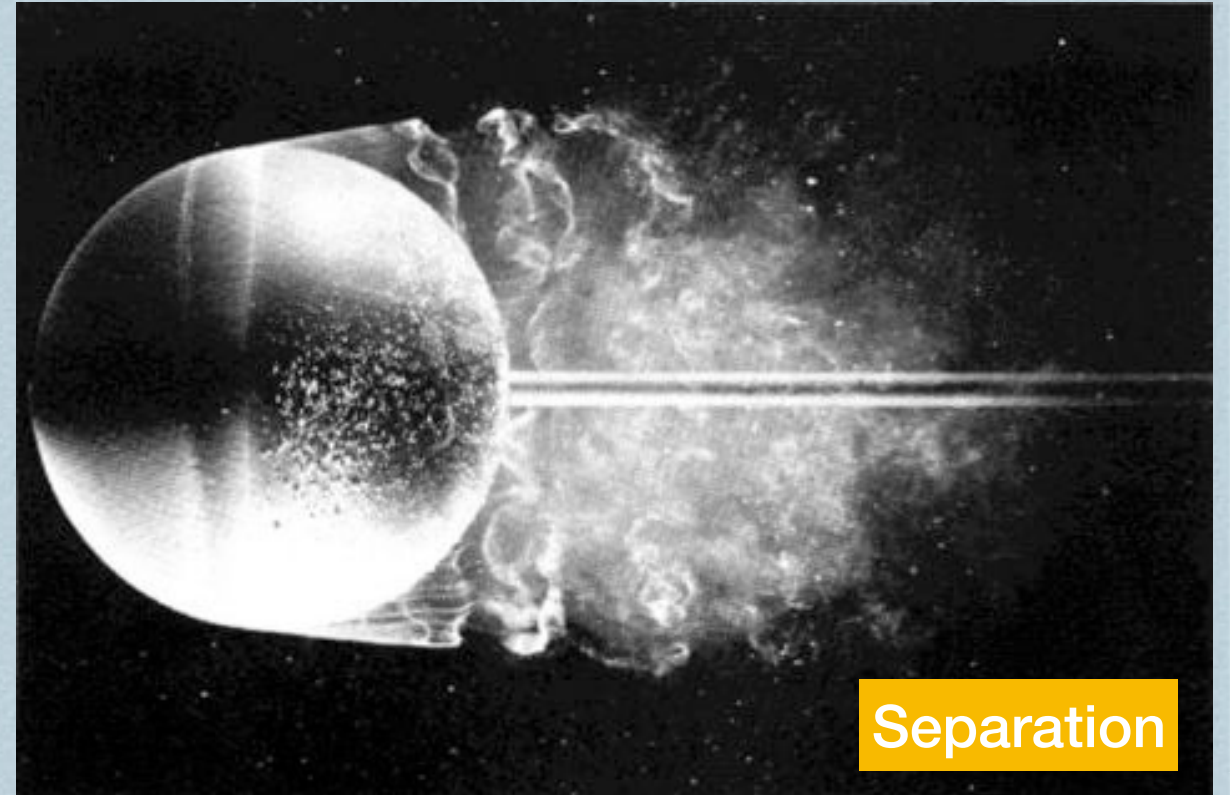
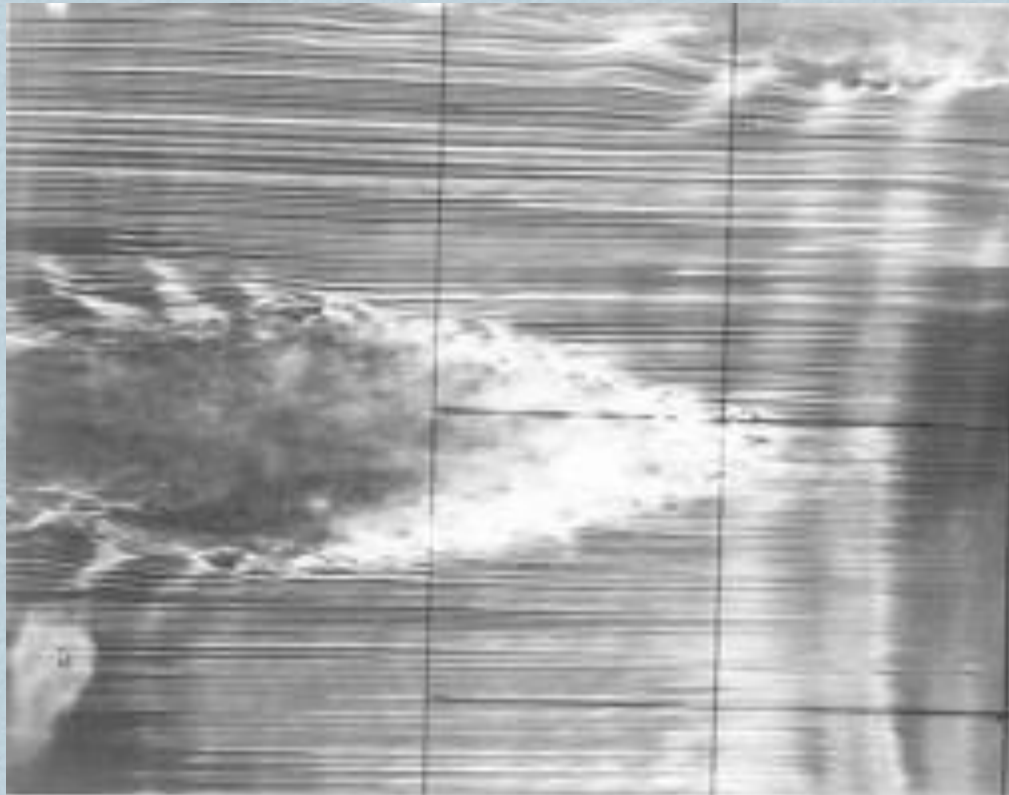
top view
of flow
from
left to
right

- ❖ Experiments suggest that viscosity matters only near solid boundaries when the flow is fast (the Reynolds number, or ratio of nonlinear to dissipative terms, is high)
- ❖ This is an idea that goes back to Ludwig Prandtl (on next slide) at the turn of the 20th century.
- ❖ In practice the majority of boundary layers are turbulent like the picture on the left.
- ❖ A typical goal in fluid mechanics courses is to derive the simplest boundary layer set of equations for laminar flow (pictured above).



Ludwig
Prandtl
in his
lab

Turbulent
spot



Separation

- From the point of view of scientific history, boundary layers are likely the pinnacle of achievement of perturbation/asymptotic theories in the 20th century.
- The boundary layer concept in fluid mechanics is both physically fundamental and of practical engineering utility.
- Based as it is on PDEs it is not, however, an ideal means to learn the mathematics behind boundary layer theory.
- The trouble is that in the context of ODEs boundary value problems are not taught as often as initial value problems.
- Let's start with a model problem that has an exact solution.

Example 8.0.1. Consider the problem

$$\begin{aligned}\epsilon y''(x) + y'(x) &= a, \\ y(0) &= 0, \\ y(1) &= 1,\end{aligned}\tag{8.0.1}$$

where $0 < a < 1$ and $0 < \epsilon \ll 1$.

The exact solution is:

$$y = ax + (1 - a) \frac{1 - e^{-x/\epsilon}}{1 - e^{-1/\epsilon}}.$$

**Notice the way that x appears with a $1/\epsilon$ coefficient.
This means instead of slow time scale we have a fast space scale.**

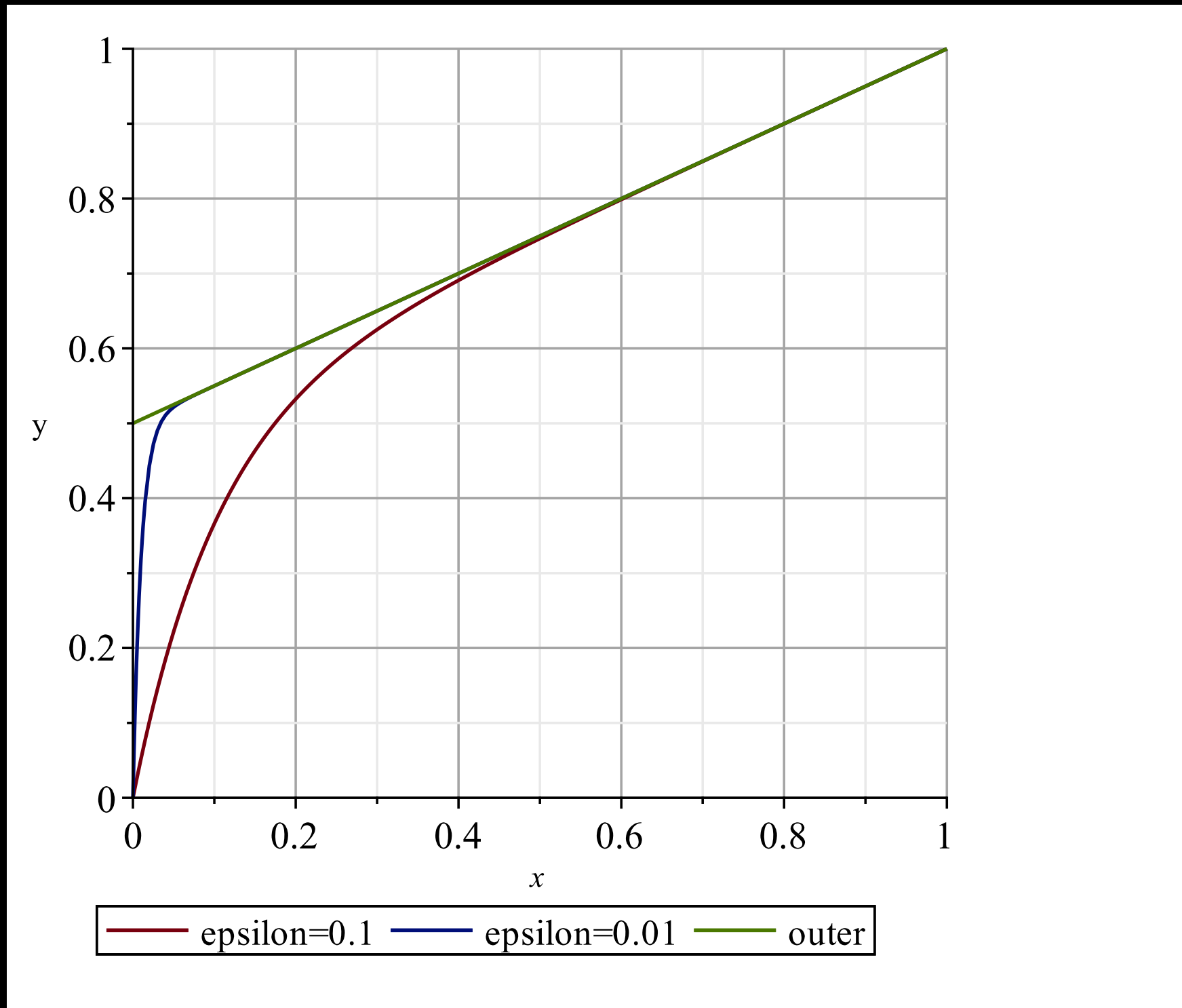
**RPT
gives**

$$y = y_0 + \epsilon y_1(x) + \epsilon^2 y_2(x) + \cdots ,$$

$$y_0' = a$$

Due to the reduced order of the PDE in RPT one cannot satisfy both BCs

Choose to satisfy the BC at $x=1$ so that $y_{outer} = 1 + a(x - 1)$



Plotting the solution, one sees that as epsilon goes to zero the outer solution is very good apart from a thin layer near the left end point

For the solution in the inner region we need to rescale x . Let

$$\chi = \frac{x}{\delta(\epsilon)}, \quad (8.0.5)$$

where $\delta(\epsilon) \rightarrow 0$ as $\epsilon \rightarrow 0$ is a measure of the currently unknown boundary layer thickness. Let $y(x) = \tilde{y}(\chi)$. In terms of $\tilde{y}(\chi)$ the ODE becomes

$$\tilde{y}'' + \frac{\delta}{\epsilon} \tilde{y}' = \frac{\delta^2}{\epsilon} a. \quad (8.0.6)$$

In the outer region the dominant balance in the differential equation was between y' and a . In the inner region the dominant balance must include $\tilde{y}''$.

Try dominant balance between the second derivative and the RHS:

$$y'' \text{ and } \frac{\delta^2}{\epsilon} a \text{ implies } \delta = \epsilon^{1/2}$$

$$\tilde{y}(\chi) = A\chi + B$$

In general both A and B are non zero and if that is so then

$$\frac{\delta}{\epsilon} y' = O(\epsilon^{-1/2})$$

But this is not smaller than the terms we kept!

The bad dominant balance (repeat last slide):

y'' and $\frac{\delta^2}{\epsilon}a$ implies $\delta = \epsilon^{1/2}$

$$\tilde{y}(\chi) = A\chi + B$$

In general both A and B are non zero and if that is so then

$$\frac{\delta}{\epsilon}y' = O(\epsilon^{-1/2})$$

But this is not smaller than the terms we kept!

The good dominant balance:

y'' and $\frac{\delta}{\epsilon}y'$ implies $\delta = \epsilon$

$$\tilde{y}'' + \tilde{y}' = \epsilon a. \quad (8.0.8)$$

We now use RPT. Setting $\tilde{y} = \tilde{y}_0 + \epsilon\tilde{y}_1 + \epsilon^2\tilde{y}_2 + \dots$, at leading order we have

$$\tilde{y}_0'' + \tilde{y}_0' = 0, \quad (8.0.9)$$

which has the general solution

$$\tilde{y}_0 = Ae^{-\chi} + B. \quad (8.0.10)$$

Which BC to apply? The one on the left from the pictures! But no general rule.

Here inner is near $x=0$ and outer is away from zero.

- (i) In the outer region: $y \approx y_{in} = y_0 = a(x - 1) + 1$.
- (ii) In the inner region: $y \approx y_{out} = \tilde{y}_0(\chi) = A(1 - e^{-\chi})$.

**What to do about the unknown constant A?
We must “match the two solutions”
or find an $\alpha > \beta > 0$ so that**

$$\begin{aligned} y &\sim y_0(x) \text{ as } \epsilon \rightarrow 0 \quad \text{for some region } x > \epsilon^\alpha, \\ y &\sim \tilde{y}_0(\chi) \text{ as } \epsilon \rightarrow 0 \quad \text{for some region } x < \epsilon^\beta. \end{aligned} \tag{8.0.12}$$

Note that $\epsilon^\beta \ll \epsilon^\alpha \ll 1$ and that both approximations are valid in the matching region $\epsilon^\beta \ll x \ll \epsilon^\alpha$.

Proving the existence of a matching region is often the most difficult part of matched asymptotic expansions. In general we want $x \rightarrow 0$ and $\chi \rightarrow \infty$ in the matching region as $\epsilon \rightarrow 0$ and we will often just assume this.

For our particular problem we can use $\epsilon \ll x \ll \epsilon^\gamma$ as the matching region for any $0 < \gamma < 1$. Taking $\gamma = 1/4$, if $\epsilon \ll x \ll \epsilon^{1/4}$ we have $y_0(x) = ax + 1 - a \sim 1 - a$ as $\epsilon \rightarrow 0$ and $\chi = x/\epsilon \gg 1$ as $\epsilon \rightarrow 0$ hence $\tilde{y}_0(\chi) = A(1 - e^{-\chi}) \sim A$ as $\epsilon \rightarrow 0$ and $\chi \rightarrow \infty$. The solutions agree iff $A = 1 - a$.

Thus we have

$$y \sim \begin{cases} a(x - 1) + 1, & \text{if } x \ll \epsilon^{1/4}; \\ (1 - a)(1 - e^{-x/\epsilon}), & \text{if } x \gg \epsilon. \end{cases} \tag{8.0.13}$$

**The domain is backwards
here and in the notes text**

- It is the matching part of the problem that is the “magnus opus”.
- Indeed in the 1970s the success of many applied math theses came down to a matching procedure.
- While in some cases this is still a worthwhile thing to do, it is a fair point to ask whether the effort is always worth it.
- Next, following the notes, I provide a different take on the procedure using a temporal boundary layer problem using the strongly damped oscillator with two measurements:

$$\frac{d^2x}{dt^2} + \frac{1}{\epsilon} \frac{dx}{dt} + x = 0 \text{ **with** } x(0) = 1, x(10) = 1/10$$

$$\frac{dx_{out}}{dt} = 0 \text{ **so that** } x_{out} = 1/10$$

The inner problem, near $t = 0$ will require us to rescale time since we expect a rapid change near this point. Let

$$\tau = \frac{t}{\epsilon^n} \quad (8.0.20)$$

where n must be chosen as part of the solution. The Chain rule gives

$$\frac{d}{dt} = \frac{d\tau}{dt} \frac{d}{d\tau} = \frac{1}{\epsilon^n} \frac{d}{d\tau}$$

so that the governing ODE for the inner domain becomes

$$\frac{1}{\epsilon^{2n}} x_{in}''(\tau) + \frac{1}{\epsilon^{n+1}} x_{in}'(\tau) + x_{in}(\tau) = 0.$$

**The boundary
layer scales with
epsilon**

It is pretty easy to see that when $n = 1$ the first two terms will balance so that $\tau = t/\epsilon$. Physically, the balance in the boundary layer is between the inertia term and the damping term. The restoring force term is not relevant. The leading order inner solution is governed by

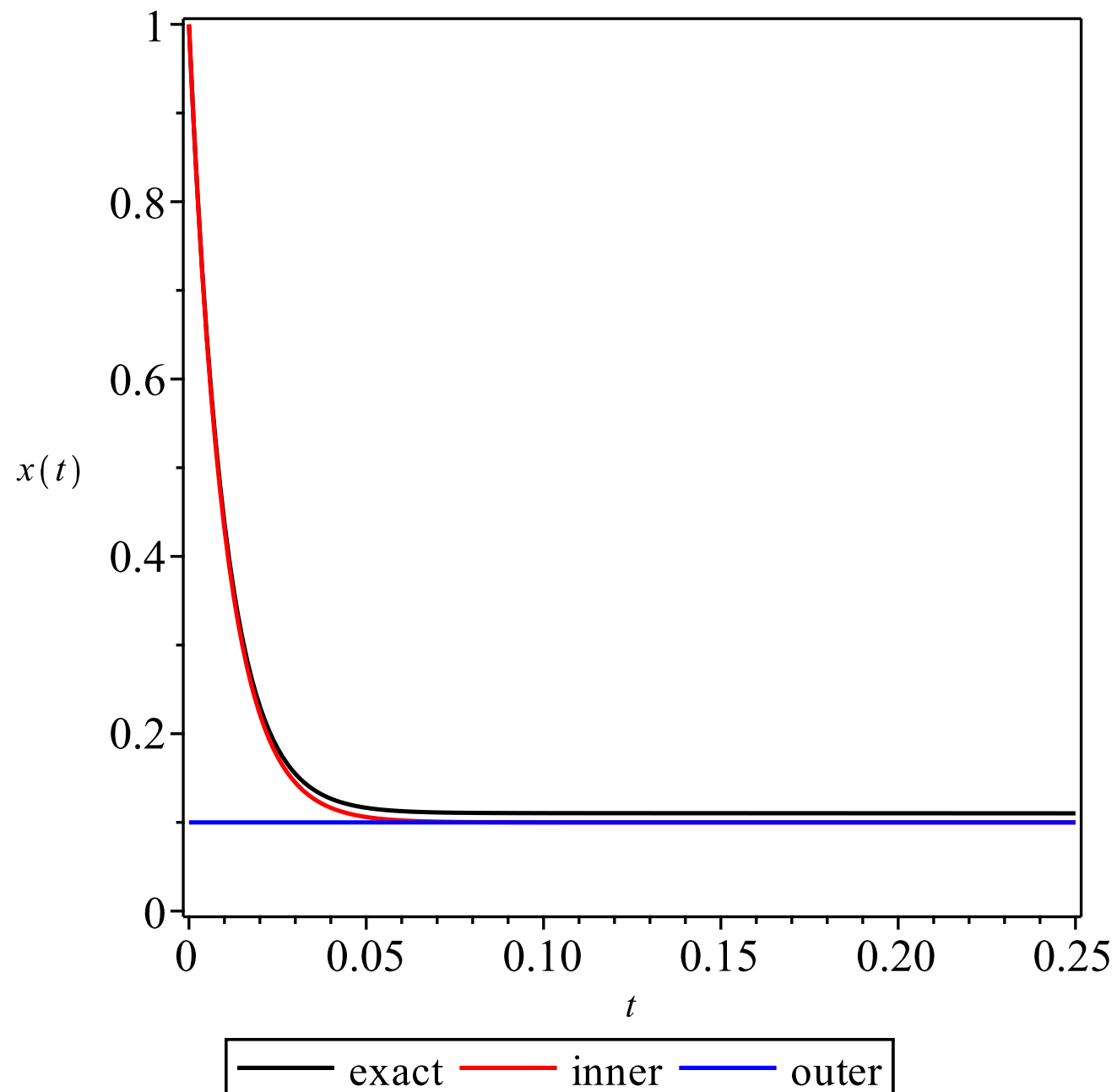
$$x_{in}''(\tau) + x_{in}'(\tau) = 0. \quad (8.0.21)$$

This is easily solved to give

$$x_{in}(\tau) = c_1 + c_2 \exp(-\tau).$$

The boundary/initial condition $x(0) = 1$ allows us to write $c_2 = 1 - c_1$ or a solution in the original variables

$$x_{in}(t) = c_1 + (1 - c_1) \exp\left(-\frac{t}{\epsilon}\right). \quad (8.0.22)$$



The actual
right
boundary is
very far to the
right

Figure 8.2: Exact and asymptotic solution of the linear BVP in Example 2 with $\epsilon = 0.01$. Only the detail of the boundary layer near $t = 0$ is shown.

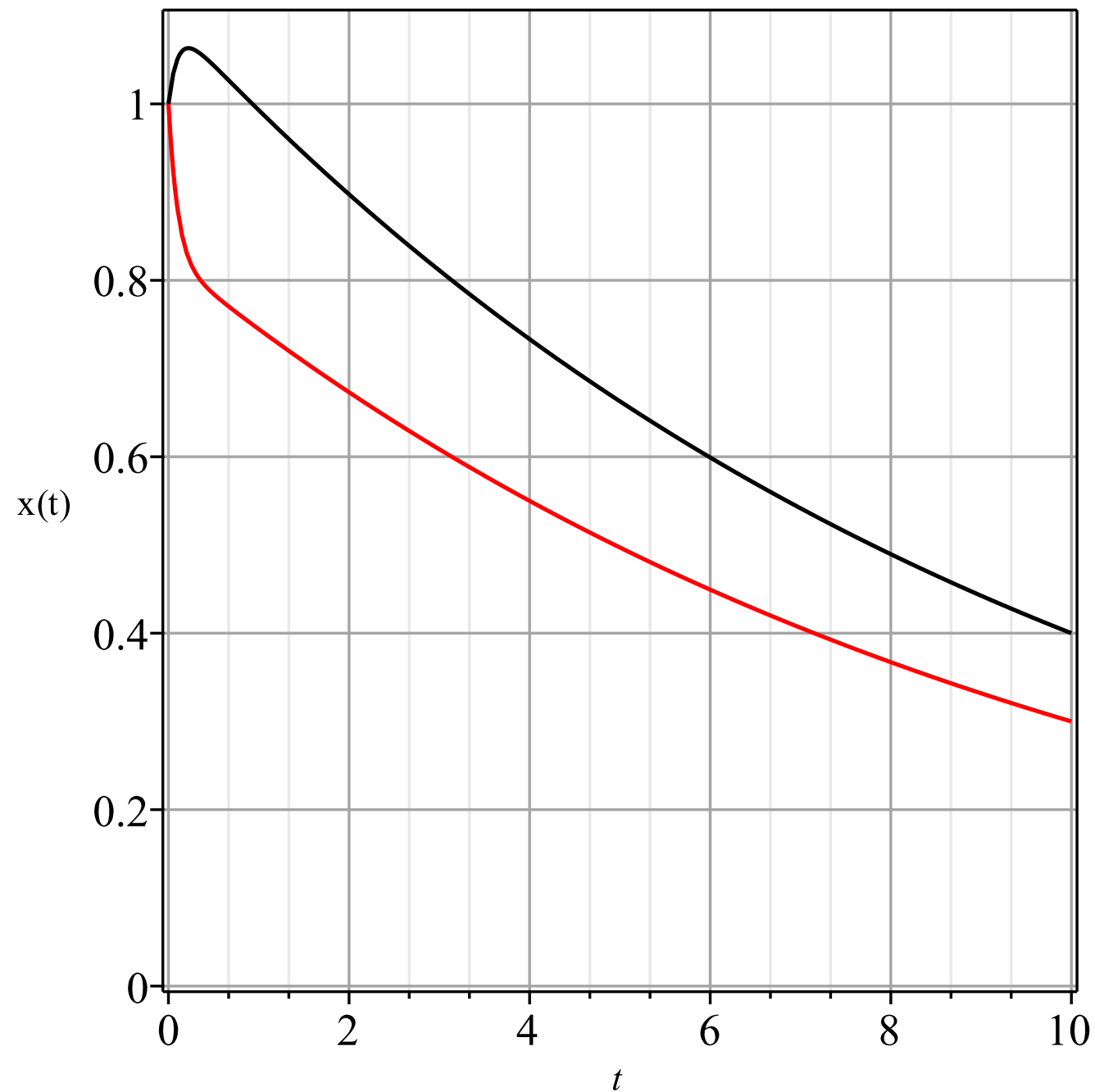
I never did do the formal matching

The formal solution of the oscillator ODE is agnostic as far as ICs or BCs are concerned. This means we can use the exact solution to solve the BVP.

$$x(t) = c_1 \exp(\lambda_1 t) + c_2 \exp(\lambda_2 t)$$

with the arbitrary constants determined by algebraic equations. For the IVP $x(0)$ sets the value of the function and $v(0)$ sets the slope of the function for early times. With the BVP however, the condition at the right hand end point sets the value of $v(0)$ implicitly. This means two things. First, we do not necessarily know where the solution is decreasing for all t in the solution interval, even when the damping is very strong. Second, we have a numerical scheme to solve the BVP using known methods for the IVP. Just set $v(0) = v_0$ and vary v_0 depending on whether a given solution is larger or smaller than the desired value at the right hand end point. Bisection is particularly easy to implement, though Newton's method is possible as well. In the literature this is often called the 'shooting method'.

But this means that the structure of the boundary layer can be influenced by the value at the far (near zero) end point.



The right boundary is the actual right boundary in this case

The change of value at $x=10$ leads to unexpected changes near $x=0$

$$\frac{d^2x}{dt^2} + \frac{1}{\epsilon} \frac{dx}{dt} + x = 0 \text{ **with** } x(0) = 1, x(10) = 1/10$$

$$\frac{dx_{out}}{dt} = 0 \text{ **so that** } x_{out} = 1/10$$

OK finally let's contrast the linearly damped SHO with a nonlinearly damped case.

$$\frac{d^2x}{dt^2} + \frac{1}{\epsilon} \left(\frac{dx}{dt} \right)^{4/3} + x = 0 \text{ **with** } x(0) = 1, x(10) = 1/10$$

$$\frac{dx_{out}}{dt} = 0 \text{ **so that** } x_{out} = 1/10$$

The inner problem is a little bit more interesting, namely

$$\frac{1}{\epsilon^{2n}} x_{in}''(\tau) + \frac{1}{\epsilon^{4n/3+1}} (x_{in}'(\tau))^{4/3} + x_{in}(\tau) = 0.$$

The first two terms balance if $2n = 4n/3 + 1$ or $n = 3/2$. Thus $\tau = t/\epsilon^{3/2}$ and the leading order inner problem reads

$$x_{in}''(\tau) + (x_{in}'(\tau))^{4/3} = 0.$$

Rewriting as a problem for $v_{in}(\tau) = x_{in}'(\tau)$ gives

$$v_{in}'(\tau) = - (v_{in}(\tau))^{4/3}$$

or

$$\frac{dv_{in}}{(v_{in}(\tau))^{4/3}} = -d\tau$$

so that integrating gives

$$\frac{3}{7} v_{in}^{7/3}(\tau) = -\tau + D_1$$

where D_1 is an arbitrary constant. Solving for v_{in} gives

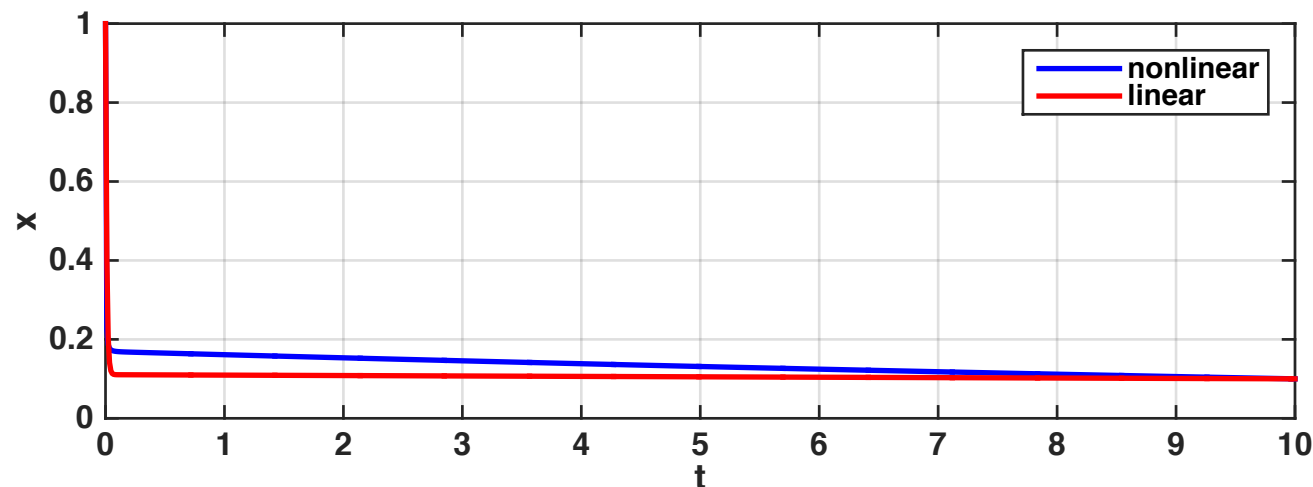
$$v_{in}(\tau) = \left(-\frac{7}{3}\tau + d_1 \right)^{3/7}$$

where I have just renamed the arbitrary constant instead of multiplying it by the factor from the LHS.

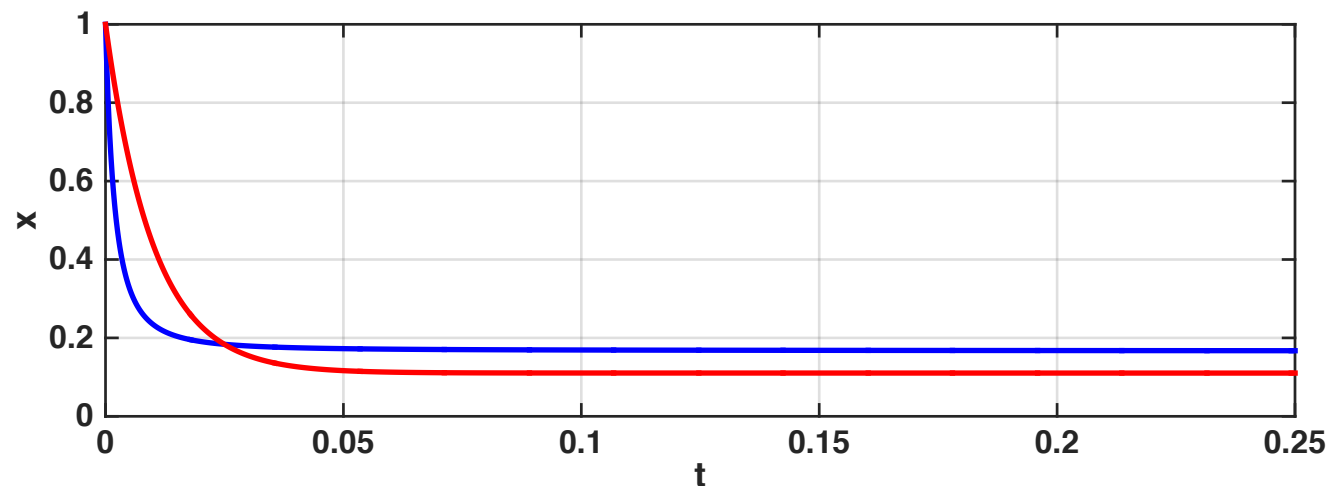
The boundary layer scales as epsilon to the 3/2

The boundary layer in the linear case scaled with epsilon

For $\epsilon=0.001$ everything behaves roughly as we think it should



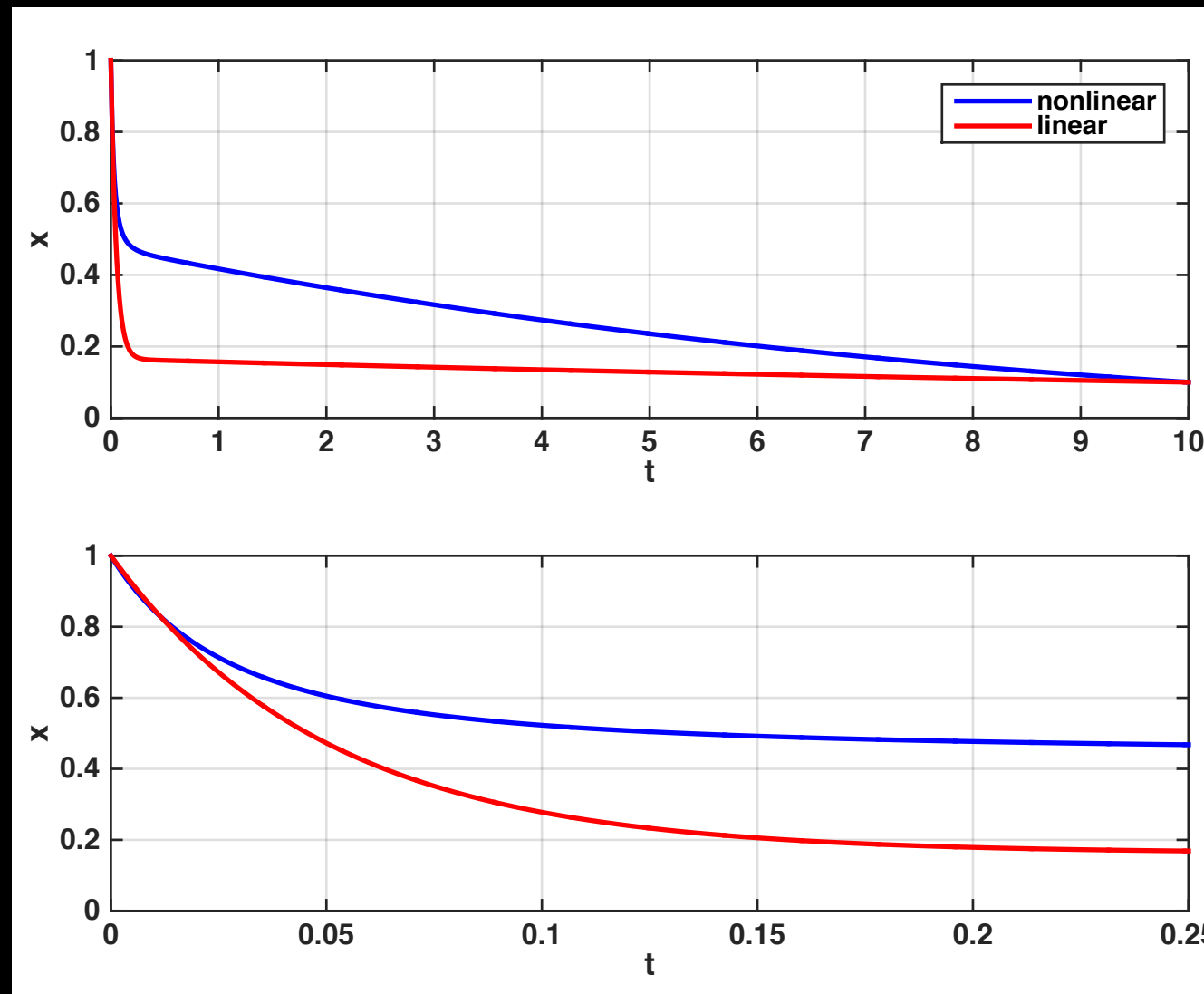
The right boundary is the actual right boundary in this case



The actual right boundary is very far to the right

The boundary layer in the linear case is thicker, and interestingly the outer solution in the nonlinear case is modified to have a nonzero slope

For $\epsilon=0.005$ you start to see difference in the nonlinear case.
In particular the balance for the outer problem is clearly not giving
a constant

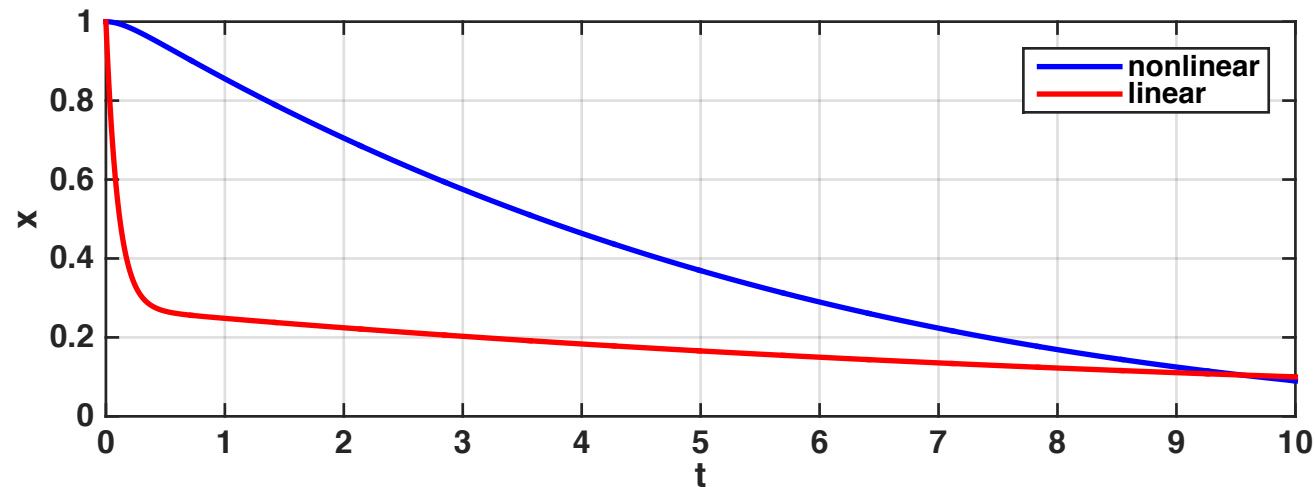


The right
boundary is
the actual
right
boundary in
this case

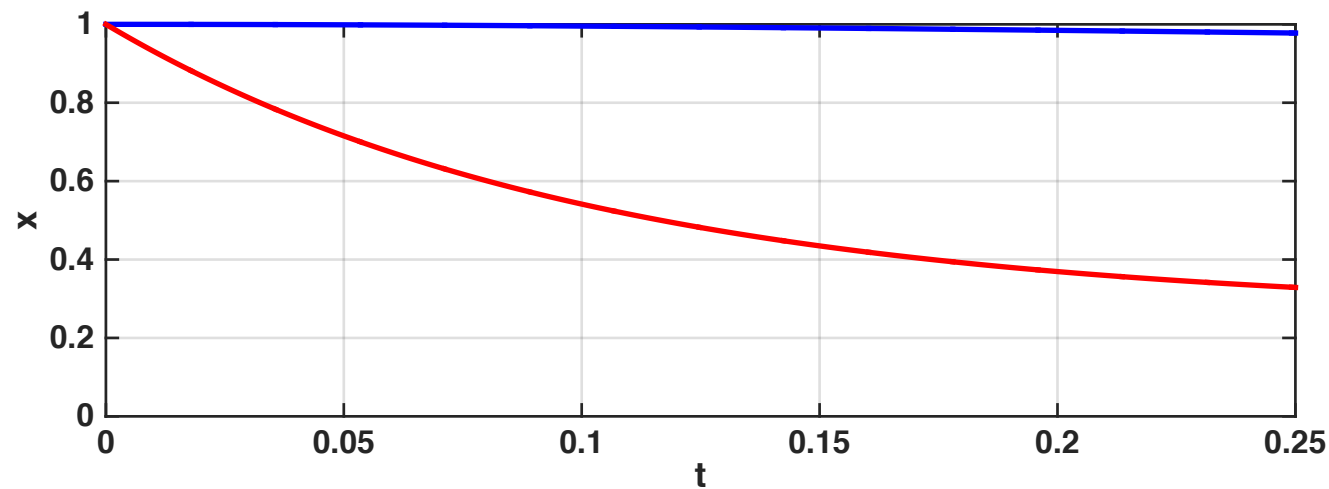
The actual
right
boundary is
very far to the
right

It is very difficult to come up with an
unambiguous comparison of the boundary layer
thickness.

For $\epsilon=0.01$ the nonlinear problem does not appear to even have a boundary layer!



The right boundary is the actual right boundary in this case



The actual right boundary is very far to the right

- We have taken a quick tour around matched asymptotic expansions.
- In only one case did we do the full treatment of finding inner and outer solutions and matching them.
- This is not to say that this is not useful, but in the spirit of this course the balance between work put in and pay off often falls short of what we would like.
- We also had a quick tour of boundary layer problems and it did not take long to uncover some seeming pathologies for both linear and nonlinear BVPs.
- We conclude with a few more facts about physical boundary layers and how they break down.

A PDE example

- Consider a 2D advection diffusion equation:

$$B_t + U_0 B_x = \kappa (B_{xx} + B_{zz})$$

- The advective part moves material left to right at a constant rate, and we ask the question of how to come up with a sensible scaling that doesn't just throw away diffusion when diffusion is weak.
- The key is to assume that typical scales in the horizontal and vertical are different: $(x, z) = (L\xi, H\zeta), H \ll L$

The dimensionless variables are thus $b(\xi, \zeta, s)$, (ξ, ζ) and s . A bit of algebra shows

$$\frac{1}{T}b_s + \frac{U_0}{L}b_\xi = \frac{\kappa}{L^2}b_{\xi\xi} + \frac{\kappa}{H^2}b_{\zeta\zeta}.$$

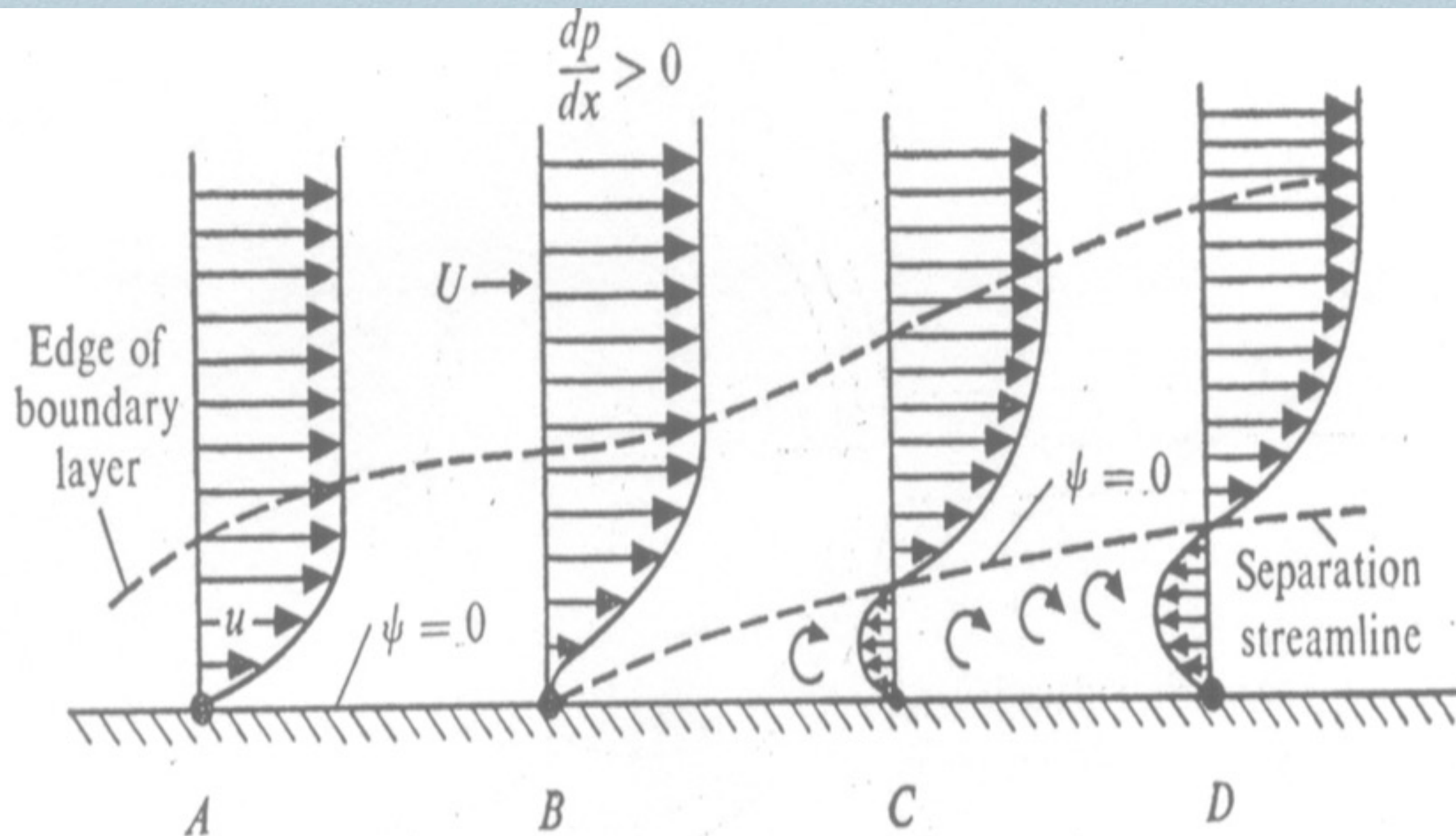
If we let $T = L/U_0$, the so-called advective time scale, then we can write

$$b_s + b_\xi = \frac{\kappa L}{U_0 H^2} \left(\frac{H^2}{L^2} b_{\xi\xi} + b_{\zeta\zeta} \right).$$

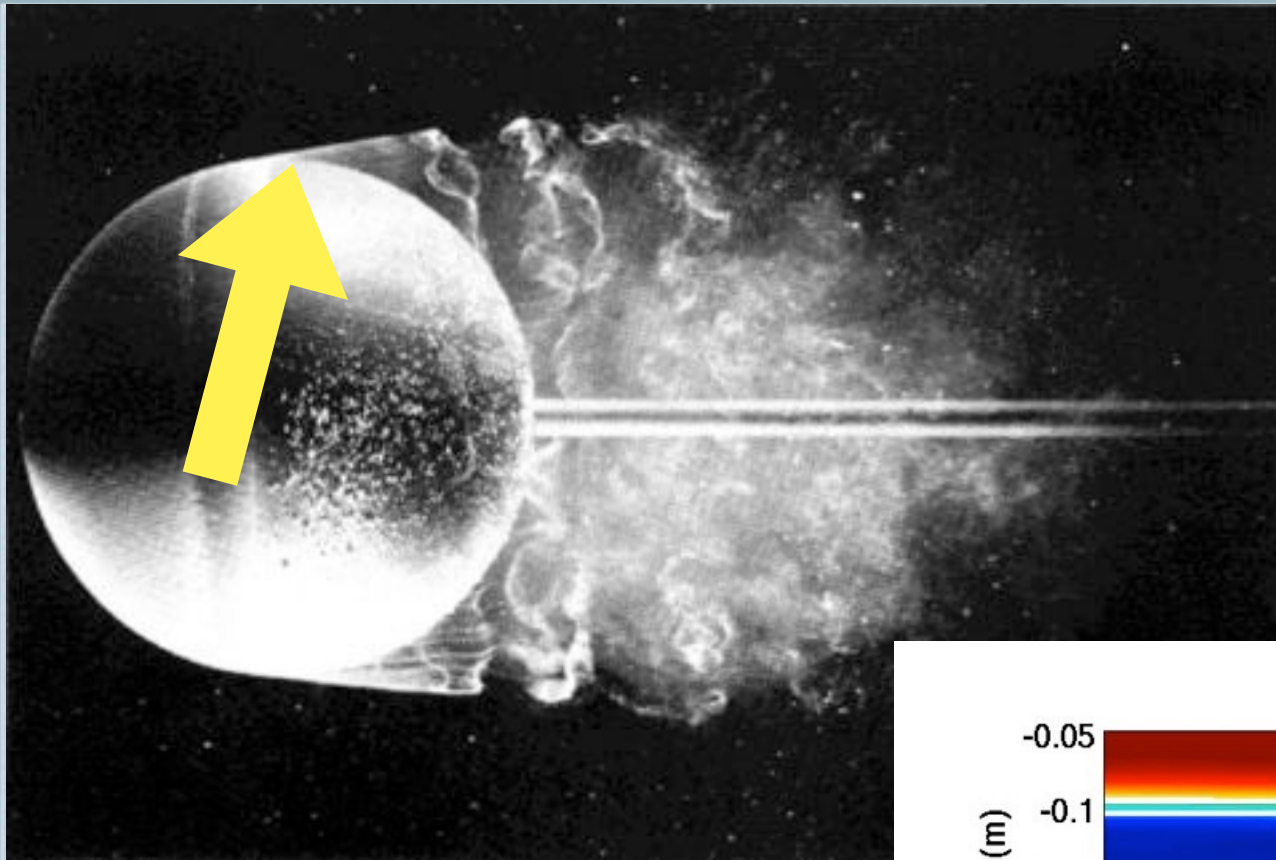
- Now define two small parameters: $\mu = \frac{H}{L}, \gamma = \frac{\kappa L}{U_0 H^2}$
- The PDE reads: $b_s + b_\xi = \gamma \left(\mu^2 b_{\xi\xi} + b_{\zeta\zeta} \right)$ meaning the Right hand side has one term that is guaranteed to be smaller than the other.
- The simplified PDE reads: $b_s + b_\xi = \gamma b_{\zeta\zeta}$

A PDE example

- The simplified PDE: $b_s + b_\xi = \gamma b_{\zeta\zeta}$ retains its highest derivative and hence can satisfy boundary conditions in the vertical direction.
- In fluid mechanics this bit of genius reconciled the observation that air and water have low viscosity with the need for the no slip boundary conditions.
- It also explained why heavier than air flight was possible.
- It is an absolute travesty that Prandtl never won the Nobel Prize in Physics!

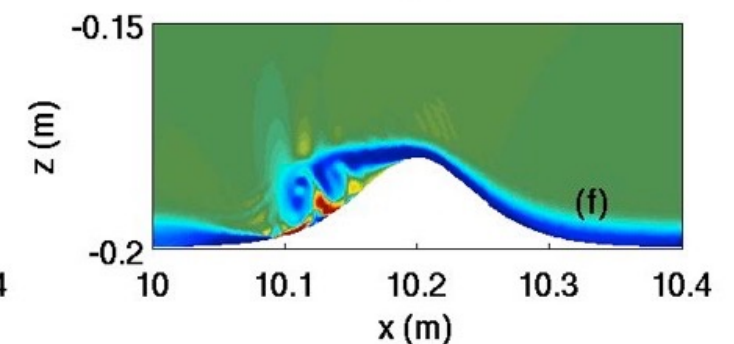
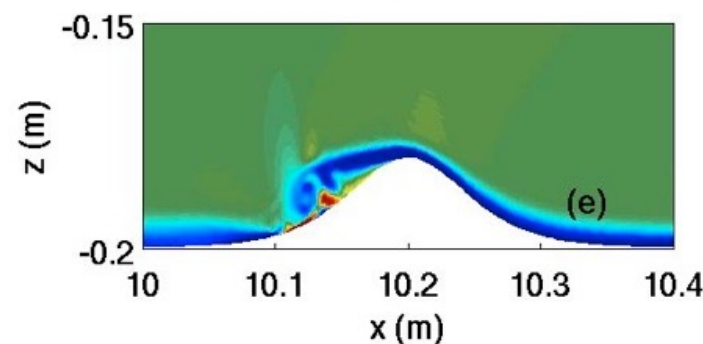
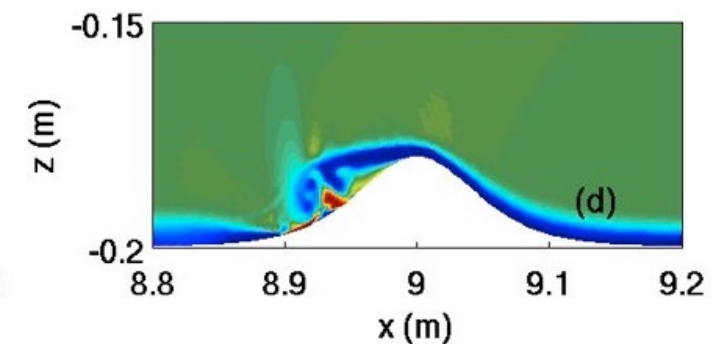
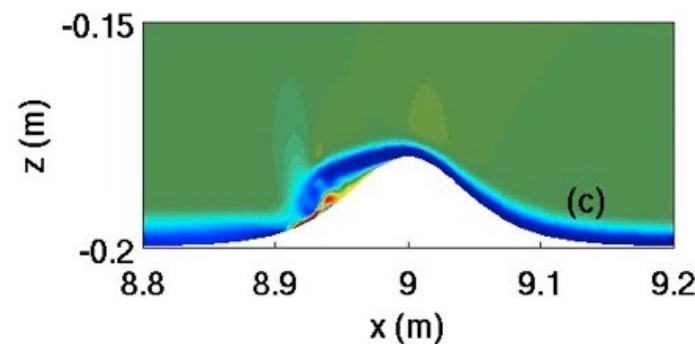
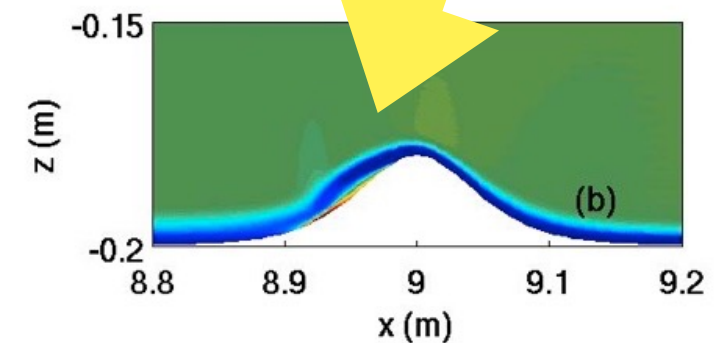
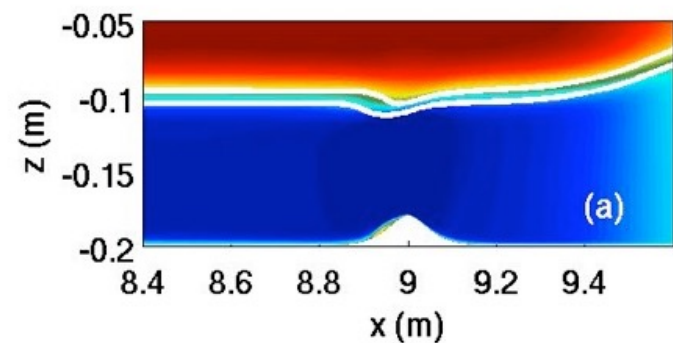


- ❖ Boundary Layers don't always stay confined to the solid boundary.
- ❖ This process is called separation and has profound effects on aerodynamics (stall), sports (curling a free kick around a wall) and other field.



❖ Separation is very difficult to address theoretically (triple deck theory).

❖ Flow separation leads to vorticity production (*top left panel shows coloured horizontal velocity*) other panels show vorticity) and breakdown of flow into turbulence



Coda: Internal Boundary Layers

- Up to now “boundary layers” have been either ODE based constructions, or something that occurred near a solid boundary.
- There are internal boundary layers that occur away from solid boundaries and these are good example of simple nonlinear PDE problems that somehow always fall through the cracks of modern PDE courses.
- I introduce two of these next.

$$A_t + AA_x = \nu A_{xx} \text{ Burgers Equation}$$

$$\xi = x - Vt \text{ travelling variable}$$

$$A(x, t) = B(\xi) \text{ travelling wave solution}$$

$$-VB' + \frac{1}{2}(B^2)' - \nu B'' = 0$$

ν is called the “viscosity”

$$-VB + \frac{1}{2}B^2 - \nu B' = C_1 \text{ integrating once}$$

- ❖ The Burgers equation is used to understand the mathematics of gas dynamics setting in a simpler context (one equation).
- ❖ First we will keep the second derivative term, then consider what happens when we drop it (for boundary layers this is really an aside).
- ❖ A basic strategy for a nonlinear equation is to look for traveling solutions (notice these can't be plane waves since Burgers equation is nonlinear).
- ❖ We want to find traveling waves that are sharp transitions between two essentially constant regions, or an internal boundary layer.

$$-VB + \frac{1}{2}B^2 - \nu B' = C_1$$

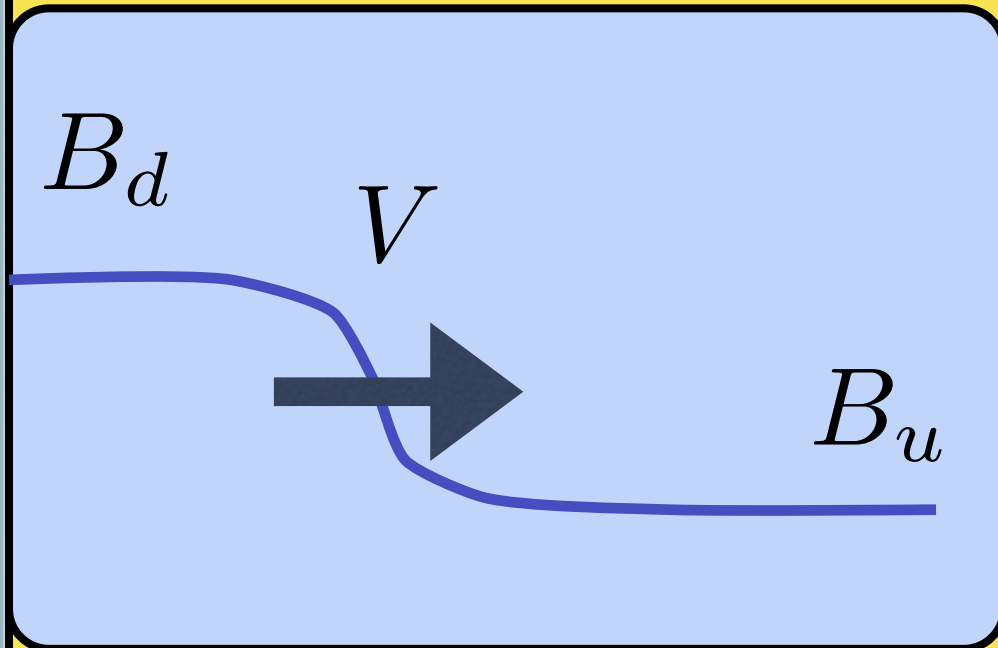
Need to find V and C_1

$$B \rightarrow B_u, B_d \text{ as } \xi \rightarrow \pm\infty$$

$$B' \rightarrow 0 \text{ as } \xi \rightarrow \pm\infty$$

$$V = \frac{1}{2}(B_u + B_d)$$

$$C_1 = -\frac{1}{2}B_u B_d$$



- ❖ We can use the upstream (or in front of shock) and downstream (or behind shock) state to integrate once and to determine the speed of the travelling disturbance and the constant of integration.
- ❖ To integrate a second time we factor and use partial fractions.

$$(B - B_u)(B_d - B) = -2\nu B' \text{ factoring}$$

$$\frac{\xi}{\nu} = \frac{2}{B_d - B_u} \log \frac{B_d - B}{B - B_u} \text{ integrating}$$

$$\Delta B = B_d - B_u \text{ define the difference}$$

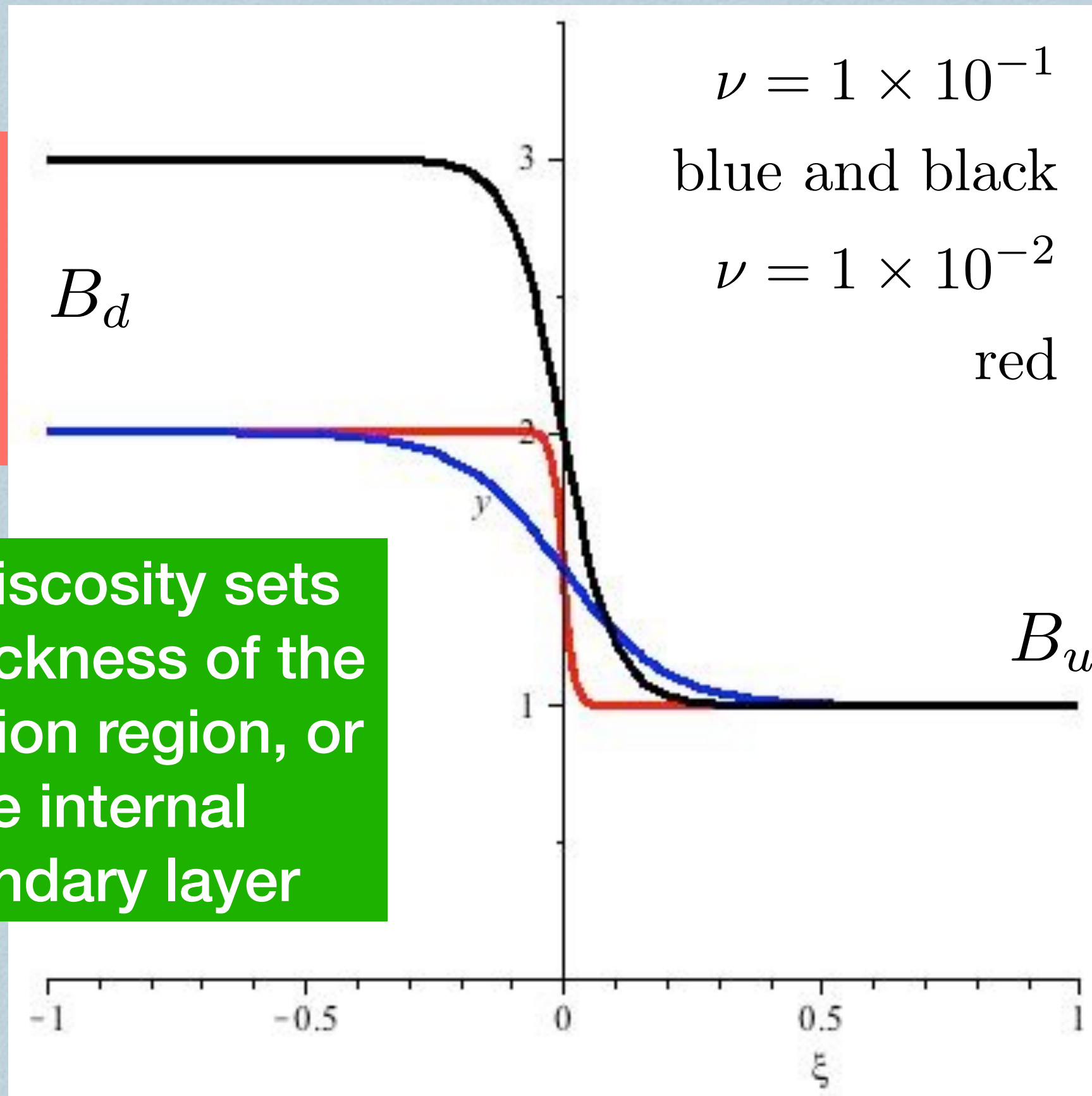
$$B(\xi) = B_u + \frac{\Delta B}{1 + \exp[\frac{\Delta B}{\nu} \xi]} \text{ final solution}$$

$$V = \frac{1}{2} (B_u + B_d)$$

- ❖ The first key point to notice is that the velocity of the travelling wave does NOT depend on the viscosity.
- ❖ The second key point to note is that the width of the wave depends inversely on viscosity, so lower viscosity means thinner waves.
- ❖ The third key point to note is that that thickness of the wave depends on the wave amplitude with larger waves being 'thinner'.

down
stream
or
behind
shock

The viscosity sets
the thickness of the
transition region, or
the internal
boundary layer



upstream
or
ahead
of
shock

- ❖ We next want to figure out what happens when the viscosity, ν , tends to zero.
- ❖ This is the same sort of limit that gives the ideal gas equations.
- ❖ From the Burgers equation solution we see that the wave width goes to zero and thus the travelling wave solution tends to a discontinuity called a **shock**.
- ❖ We want to use some PDE theory to understand these solutions.
- ❖ The simplest problem is called the Riemann problem.

$A_t + AA_x = 0$ inviscid Burgers equation

$$A(x, 0) = A_d > A_u \text{ when } x < 0$$

$$A(x, 0) = A_u > 0 \text{ when } x > 0$$

- ❖ We could define a characteristic curve in analogy with the ξ variable in the travelling wave solution for the viscous Burgers equation:

$$\frac{dx}{dt} = A(x, t)$$

$$\frac{dx}{dt} = A(x, t)$$

- ❖ But for the Riemann problem on the last slide we have that

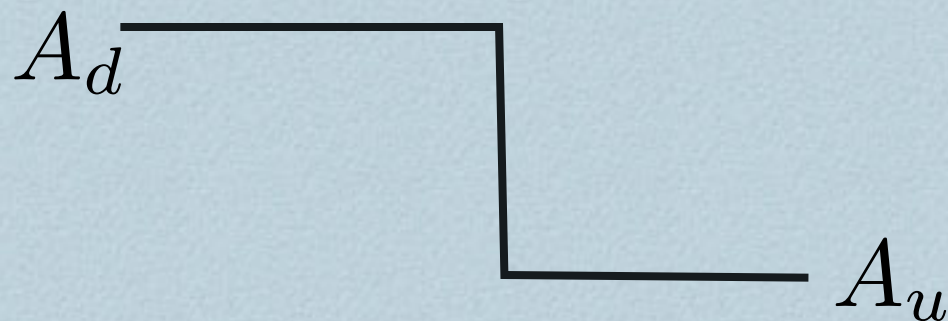
$$\frac{dx}{dt} = A_d \text{ when } x < 0$$

$$\frac{dx}{dt} = A_u \text{ when } x > 0$$

But $A_d > A_u$ so the curves cross

- ❖ Another way to see this is to write down an implicit solution using D'Alembert's solution for the advection equation ($F(\cdot)$ is given by the initial condition):

$$A(x, t) = F[x - A(x, t)t]$$

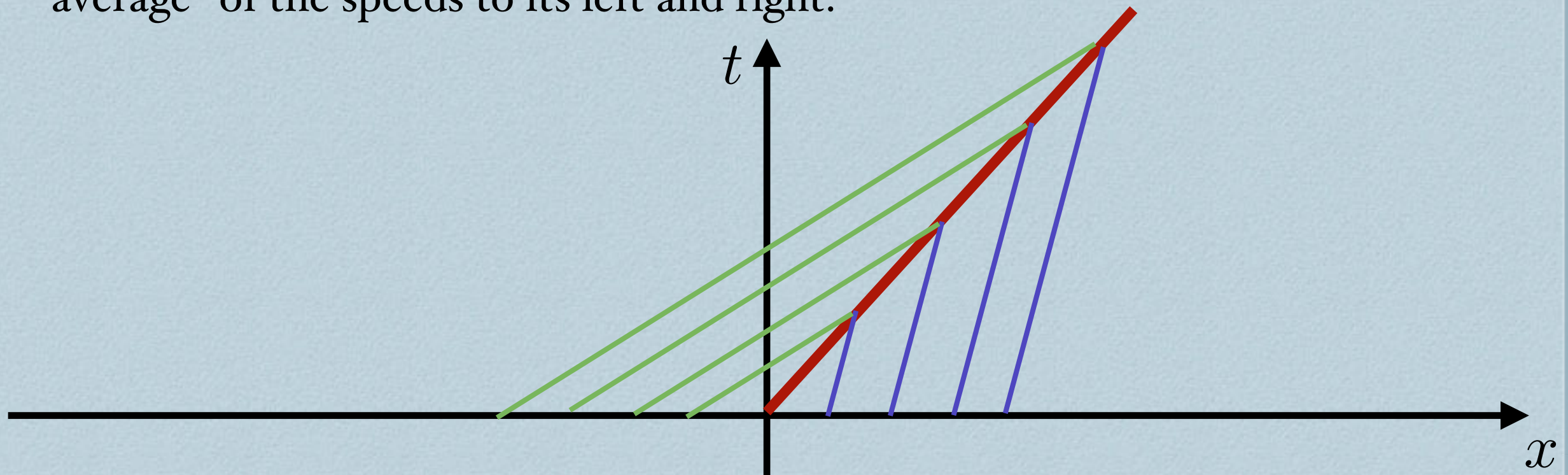


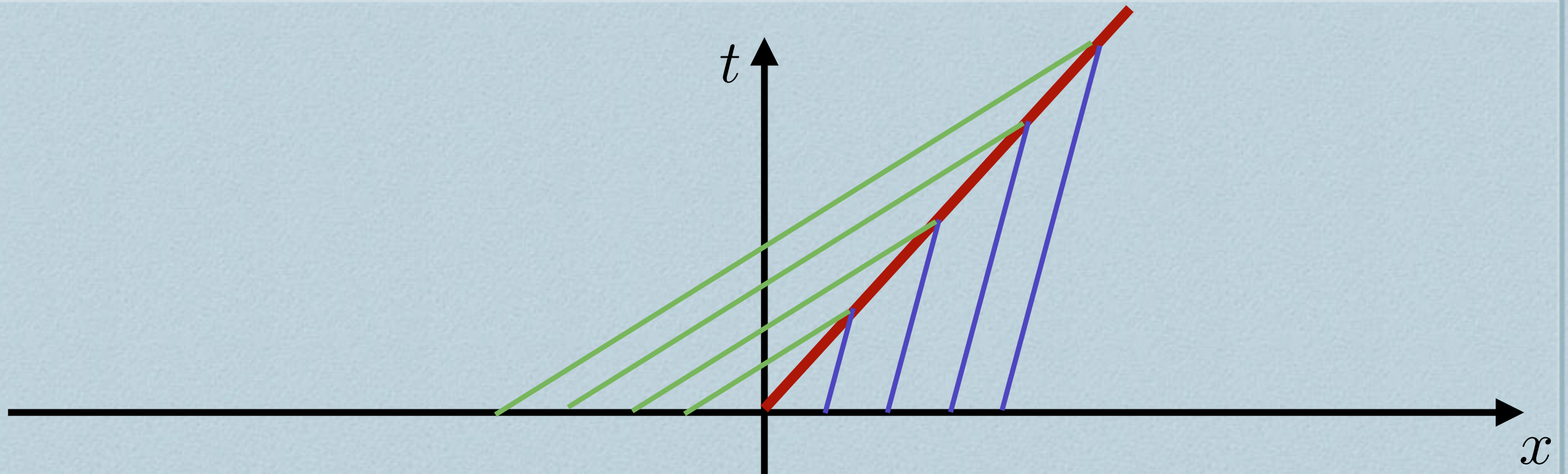
- ❖ So the implicit solution loses uniqueness immediately!

- ❖ How can we fix things? Let's return to the Burgers equation with viscosity.
- ❖ We found for this case that the wave speed did not depend on viscosity so take the wave speed to match the viscous case and solve the initial value problem as:

$$A(x, t) = A_d + (A_u - A_d) H \left(x - \frac{A_u + A_d}{2} t \right)$$

- ❖ Where $H(\cdot)$ is the Heaviside step function.
- ❖ All that remains to do is to fix the characteristic curves.
- ❖ To do this start with the discontinuity. At $t=0$ it is at $x=0$ and moves with the “average” of the speeds to its left and right.





- ❖ The steeper characteristics that start from the positive x axis are shown in blue.
- ❖ The less steep characteristics that start from the negative x axis are shown in green.
- ❖ The path the shock follows is shown in red. Shocks are thus the limit of infinitely thin boundary layers.
- ❖ Returning to the original PDE you might notice that we have gone slightly out of bounds from a mathematical point of view.
- ❖ This is because the shock solution is not differentiable (it isn't even continuous!).

Reaction Diffusion

- The Burgers equation is a classical example of a nonlinear, scalar PDE.
- Other examples of similar importance are the KdV equation for nonlinear, dispersive waves and Fisher's equation.
- The latter exhibits internal boundary layers (and shows not all cool PDEs come from physics) so we will consider it next.

$$B_t = \nu B_{xx} + rB(1 - B)$$

Fisher's Equation

$$A_t = A_{xx} + A(1 - A)$$

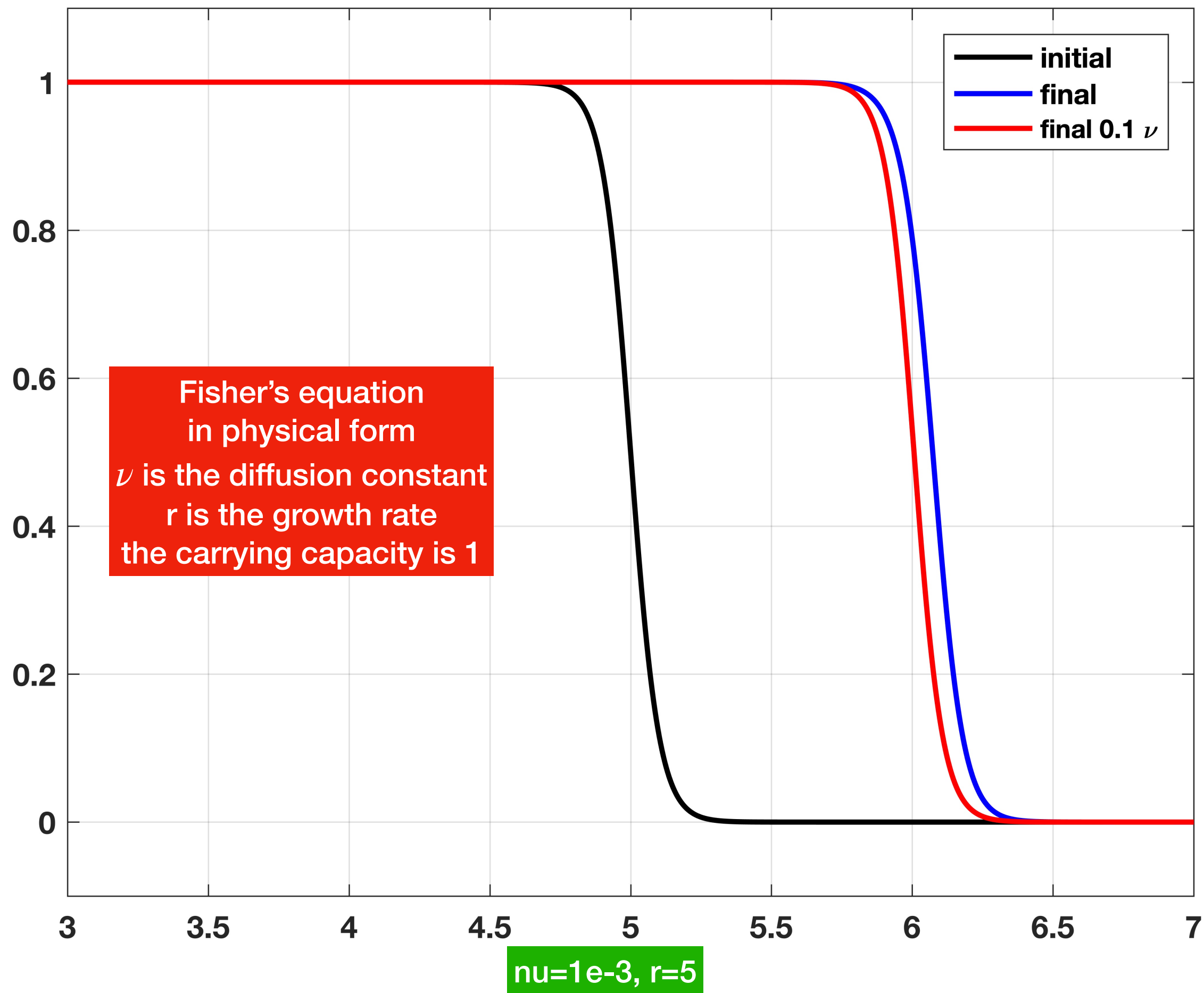
- Fischer's equation (dimensionless form given above) has traveling wave solutions, but not in closed form.
- The way to prove it has exact solutions is interesting, but it is beyond AMATH 732 course material. See Logan's "An Introduction to Nonlinear Partial Differential Equations" for details.
- The idea is to look for traveling wave solutions by switching to the variable: $\xi = x - Vt$.
- This gives a 2 dimensional dynamical system with two fixed points. One fixed point is a stable focus, the other a saddle point.

Fisher's Equation: $A_t = A_{xx} + A(1 - A)$

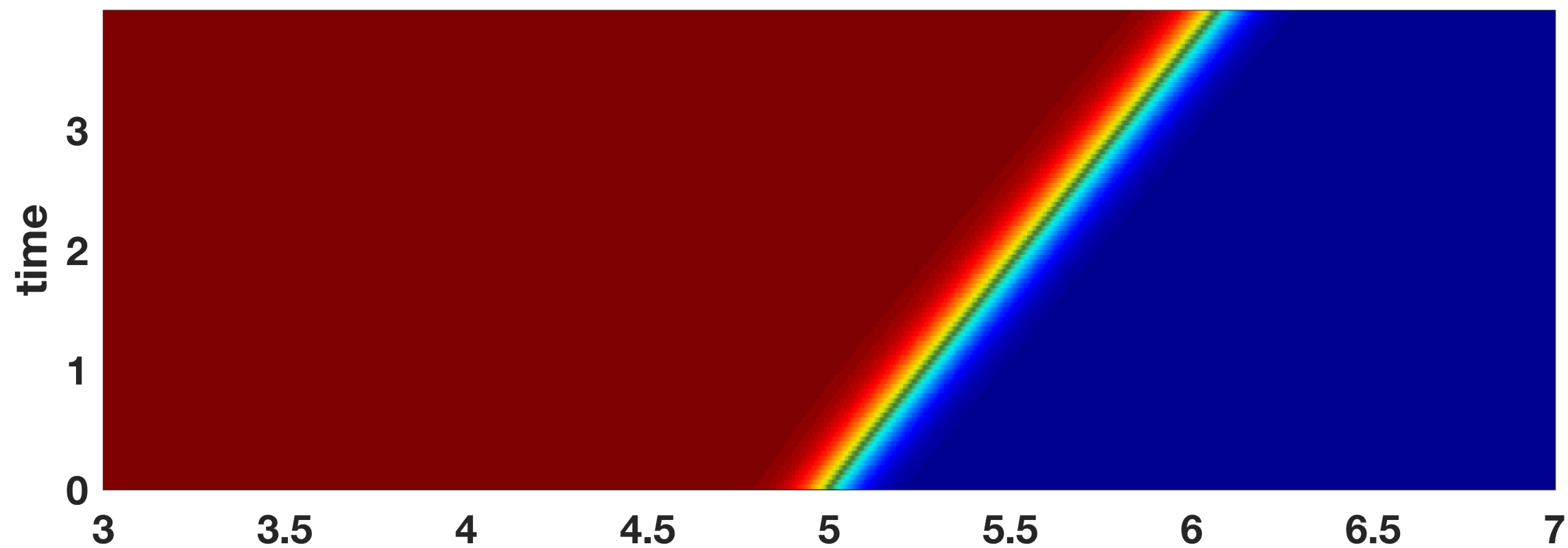
- Fischer's equation (dimensionless form given above) has traveling wave solutions, but not in closed form.
- The way to prove it has exact solutions is interesting, so I will post the proof, but it is beyond AMATH 732 course material.
- The idea is to look for traveling wave solutions by switching to the variable: $\xi = x - Vt$.
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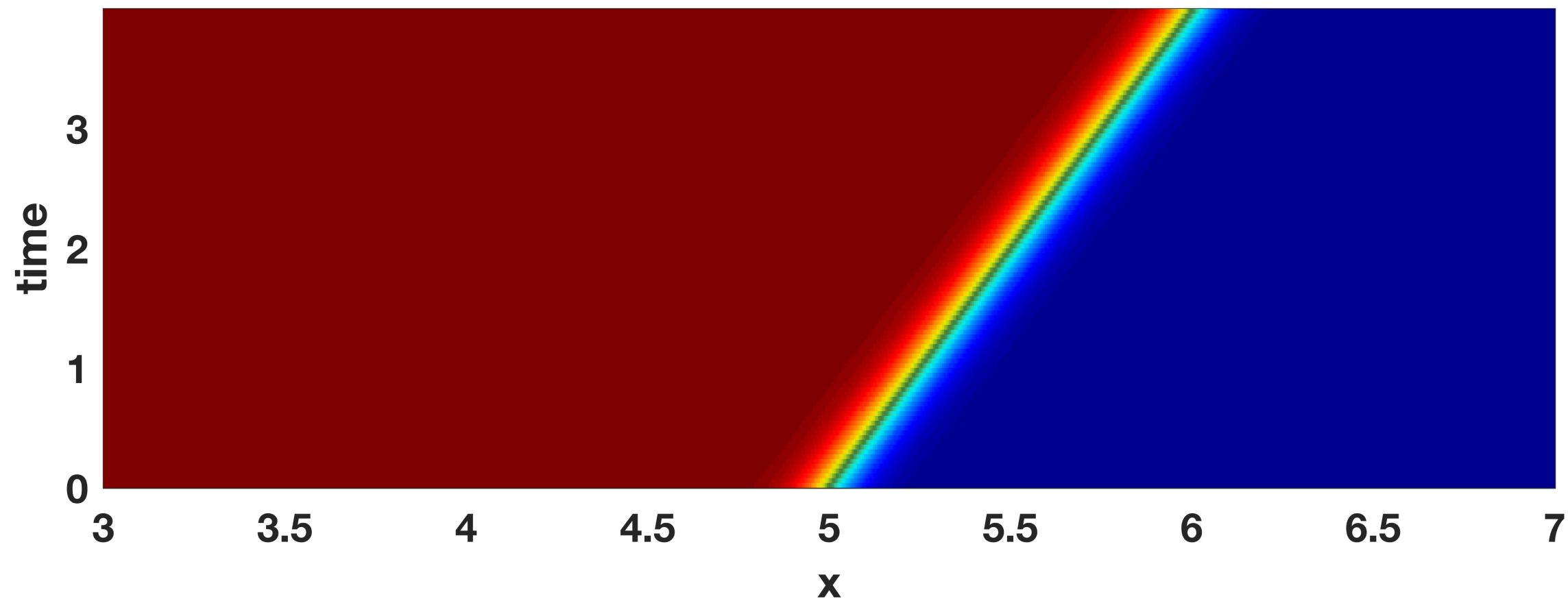
- The technical aspect of the proof comes down to proving the existence of a heteroclinic orbit (one that links the two fixed points).
- Mathematicians often stop after answering the “is it possible question?”
- But the real question is do traveling wave solution arise spontaneously?
- I tested this with an FFT based Matlab code.



$\nu=1e-3, r=5$



$\nu=1e-4, r=5$



Summary

- Internal boundary layers are spontaneously arising features of a variety of simple, nonlinear PDEs.
- Typically they arise because of a combination of systems.
- They are sometimes referred to as traveling waves.
- This is a slight misnomer since there is no restoring force in many of these models and the correct analogy is something like a forest fire which spreads in space leaving behind burnt ground that cannot burn again.